



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 19 Optimization V: Practice Questions

- The goal of today's lecture is to guide you through a few questions and reinforce your understanding of *how to prove a convex set*.

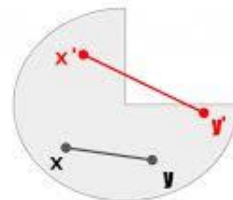
Review: How to Check a Set is Convex?

You can consider using:

- **Visualization** – If the set is in 1D or 2D, we can use a geometric approach to check convexity.



Convex Set



Non-Convex Set

- **Definition-based proof** – A set C is convex if $\forall x_1, x_2 \in C, \forall \theta \in [0,1]$

$$\theta x_1 + (1 - \theta)x_2 \in C.$$

Review: How to Check a Set is Convex?

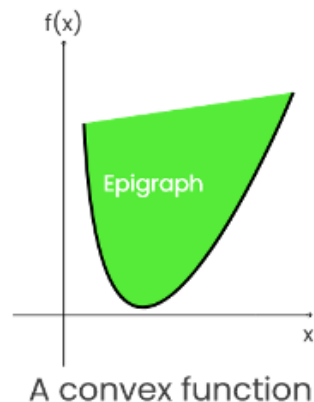
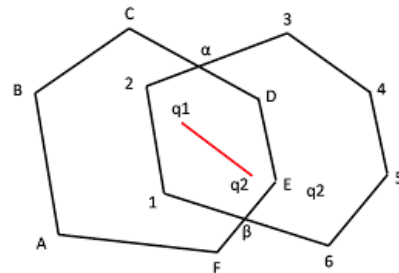
You can consider using:

- **Intersection property** – The intersection of convex sets is always convex

- If the set is an **epigraph** of $f(x)$, that is

$$\mathbf{epi}(f) = \{(x, t) \mid t \geq f(x)\}$$

then suppose f is convex, its epigraph is a convex set.



Review: Typical Examples of Convex Sets

- The empty set $\emptyset$, the singleton set $\{\mathbf{x}_0\}$, and the complete space R are convex sets.
- An interval of $[a, b] \subset R$ is a convex set
- In R^n the set $H := \{\mathbf{x} \in R^n: a_1x_1 + \cdots + a_nx_n = c\}$ is a convex set
- Half spaces, e.g., $H := \{(x, y): y \leq ax + b\}$ are convex sets
- A disk with center $(0,0)$ and radius c is a convex subset of R^2

Consider using the definition or visualization to prove the above results.

Outline

Exercise 1 (Taken from Assignment 3)

Exercise 2 (Taken from Assignment 3)

Exercise 3 (Taken from Assignment 3)

Exercise 4 (Last Year's Final Exam)

Outline

Exercise 1 (Taken from Assignment 3)

Exercise 2 (Taken from Assignment 3)

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Exercise 4 (Last Year's Final Exam)

Exercise 1:

- **Question:** Define the square (in $\mathbf{R}^2$) $S = \{\mathbf{x} = (x_1, x_2) \in \mathbf{R}^2 : 0 \leq x_i \leq 1, i = 1, 2\}$, and the disk $D = \left\{ \mathbf{x} = (x_1, x_2) \in \mathbf{R}^2 : \sqrt{(x_1)^2 + (x_2)^2} \leq 1 \right\}$. Determine whether the following statements are true or false. Hint: you can use figures to show the results.
- (a). $S \cap D$ is convex.
 - (b). $S \cup D$ is convex.
 - (c). $S \cap D^C$, i.e. S/D is convex.

This question is taken from our assignment 3

Solution to Exercise 1

- ▶ **Solution:** You can draw some figures for S and D to understand their definitions.
 - ▶ (a). True
 - ▶ (b). True.
 - ▶ (c). False.

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Exercise 4 (Last Year's Final Exam)

Exercise 2

- ▶ **Question:** Determine whether the following statements are true or false.
 - ▶ (a). $S = \{(x, y, z) : z \geq x^2 + y^2\} \subset \mathbb{R}^3$ is convex;
 - ▶ (b). The intersection of two non-convex sets **could be** a convex set.

This question is taken from our assignment 3

Solution to Exercise 2

- **Solution:** (a). True. Assume that $a_1 = (x_1, y_1, z_1)$, $a_2 = (x_2, y_2, z_2)$ are both in S , so that $z_1 \geq x_1^2 + y_1^2$, $z_2 \geq x_2^2 + y_2^2$. Convexity of S is the same as showing, for $0 \leq \lambda \leq 1$, that,

$$(1 - \lambda)a_1 + \lambda a_2 = ((1 - \lambda)x_1 + \lambda x_2, (1 - \lambda)y_1 + \lambda y_2, (1 - \lambda)z_1 + \lambda z_2) \in S$$

i.e., that,

$$((1 - \lambda)x_1 + \lambda x_2)^2 + ((1 - \lambda)y_1 + \lambda y_2)^2 \leq (1 - \lambda)x_1^2 + \lambda x_2^2$$

(Since this can be reorganized to $\lambda(1 - \lambda)(x_1^2 + x_2^2 - 2x_1x_2) = \lambda(1 - \lambda)(x_1 - x_2)^2 \geq 0$. Convexity of the function $f(x) = x^2$ is really what is at play here. We will return to this later), and similarly

$$((1 - \lambda)y_1 + \lambda y_2)^2 \leq (1 - \lambda)y_1^2 + \lambda y_2^2$$

Solution to Exercise 2

► **Solution (Continued):** It follows that

$$\begin{aligned} ((1 - \lambda)x_1 + \lambda x_2)^2 + ((1 - \lambda)y_1 + \lambda y_2)^2 &\leq (1 - \lambda)x_1^2 + \lambda x_2^2 + (1 - \lambda)y_1^2 + \lambda y_2^2 \\ &= (1 - \lambda)(x_1^2 + y_1^2) + \lambda(x_2^2 + y_2^2) \\ &\leq (1 - \lambda)z_1 + \lambda z_2 \end{aligned}$$

(b). True. The intersection of two non-convex sets could still be non-convex or empty. In general, the intersection of two non-convex sets does not guarantee convexity.

Outline

Exercise 1 (Taken from Assignment 3)

Exercise 2 (Taken from Assignment 3)

Exercise 3 (Taken from Assignment 3)

Exercise 4 (Last Year's Final Exam)

Exercise 3:

- ▶ For $x = [x_1, x_2] \in \mathbb{R}^2$,
 - ▶ (a) Show the convexity of $S_1 = \{x \in \mathbb{R}^2 \mid a_1 x_1 + a_2 x_2 \leq \beta\}$
 - ▶ (b) Show the convexity of $S_2 = \{x \in \mathbb{R}^2 \mid \alpha \leq a_1 x_1 + a_2 x_2 \leq \beta\}$

Solution to Exercise 3:

- (a). For $(x_1, x_2) \neq (y_1, y_2) \in S, \forall \theta \in [0, 1]$, we have

$$\begin{aligned} & a_1 [\theta x_1 + (1 - \theta)y_1] + a_2 [\theta x_2 + (1 - \theta)y_2] \\ &= \theta [a_1 x_1 + a_2 x_2] + (1 - \theta) [a_1 y_1 + a_2 y_2] \\ &\leq \theta \beta + (1 - \theta) \beta \\ &= \beta \end{aligned}$$

where we apply the fact that $(x_1, x_2) \neq (y_1, y_2) \in S$. Thus, S_1 is a convex set.

- (b). Similarly, $S_3 = \{x \in \mathbb{R}^2 \mid a_1 x_1 + a_2 x_2 \geq \alpha\}$ is a convex set. Then $S_2 = S_1 \cap S_3$ is a convex set.

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Exercise 1 (Taken from Assignment 3)

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Exercise 4 (Last Year's Final Exam)

Exercise 4:

► (15 points) **Convex Sets**

- (a) (5 points) Let $f(x)$ be a convex function and S a convex set. Then the optimization problem $\min_{x \in S} f(x)$ is a convex optimization problem. Let $C \subseteq S$ be the set of points such that $\mathbf{x} \in C$ if and only if x is an optimal solution to the problem. Show that C is a convex set.
- (b) (5 points) Show that the set S given by $S = \{(x, y) \mid x^2 \leq 5 - 2y^2\}$ is a convex set.
- (c) (5 points) Let $S_1 \subseteq \mathbb{R}^n$ and $S_2 \subseteq \mathbb{R}^n$ be two convex sets. Show that $S = \{\mathbf{x} + \mathbf{y} \mid \mathbf{x} \in S_1, \mathbf{y} \in S_2\}$ is a convex set.

This question is taken from last year's final exam.

Solution to Exercise 4

- (a) Given any two points $\mathbf{x}$ and $\mathbf{y}$ such that $\mathbf{x} \in C$ and $\mathbf{y} \in C$. For any $\lambda \in [0, 1]$, we can obtain that

$$\begin{aligned} f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) &\leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}) \\ &= f(\mathbf{x}) \end{aligned}$$

where the equality holds because both $\mathbf{x}$ and $\mathbf{y}$ are optimal solutions to the problem. Hence, $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y})$ also achieves the minimum and $\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in C$ which completes the proof.

Solution to Exercise 4

- ▶ (b) In the class, we have shown that S is the sub-level set of a convex function $f(x, y) = x^2 + 2y^2$ and therefore is a convex set.

Solution to Exercise 4

- (c) Given any two points $\mathbf{x}, \mathbf{z} \in S_1$ and other two points $\mathbf{y}, \mathbf{w} \in S_2$, for any $\lambda \in [0, 1]$, we can obtain that

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in S_1, \text{ and}$$

$$\lambda \mathbf{z} + (1 - \lambda) \mathbf{w} \in S_2$$

Therefore,

$$\lambda(\mathbf{x} + \mathbf{y}) + (1 - \lambda)(\mathbf{z} + \mathbf{w}) = (\lambda \mathbf{x} + (1 - \lambda) \mathbf{z}) + (\lambda \mathbf{y} + (1 - \lambda) \mathbf{w}) \in S.$$